Shobhit S. Chaturvedi

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RESEARCH INTERESTS & VISION

I am a computational chemist investigating how **protein scaffolds**—particularly **local electric fields** and **protein dynamics**—shape catalysis and guide evolution. My vision is to integrate **data-driven approaches** (e.g., machine learning) with **multiscale simulations** (QM/MM, MD) on high-performance computing platforms to both elucidate **fundamental mechanisms** of biocatalysis and **engineer enzymes** with novel or enhanced functions. By **capturing and quantifying** how the protein's scaffold facilitates catalysis, I seek to **enable rational enzyme design**—whether by **reengineering proteins for new-to-nature catalysis** or achieving **de novo designs** for biocatalysis, biotechnology, and therapeutic applications.

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar
Department of Chemistry and Biochemistry, University of California, Los Angeles — January 2023 – Present
Advisor: Dr. Anastassia N. Alexandrova

Graduate Research Assistant
Department of Chemistry, Michigan Technological University — May 2019 – December 2022
Advisor: Dr. Christo Z. Christov

Graduate Teaching Assistant
Department of Chemistry, Michigan Technological University — January 2018 - May 2019
Advisor: Dr. Christo Z. Christov

Project Research Assistant
Department of Chemical Engineering, Indian Institute of Technology Bombay — June 2017 - December 2017
Advisor: Dr. Ateeque Malani

Summer Intern
Department of Chemical Engineering, Indian Institute of Technology Bombay — May 2016 - June 2016
Advisor: Dr. Ateeque Malani

EDUCATION

Doctor of Philosophy in Chemistry
Michigan Technological University, — January 2018 – December 2022
Advisor: Dr. Christo Z. Christov

Master of Science in Chemistry
Michigan Technological University, — January 2018 – December 2022
Advisor: Dr. Christo Z. Christov

Bachelor of Technology in Chemical Engineering
Rashtrasant Tukadoji Maharaj Nagpur University, — June 2014 - May 2017

RESEARCH EXPERTISE

Machine Learning & Data-Driven Approaches

- Integrated machine learning with atomistic simulations to predict protein function from electric-field "fingerprints."
- Created automated workflows (Bash/Python/HPC pipelines) for big-data analysis, accelerating enzyme screening and design.

Dynamic 3D Electric Field Mapping

- Pioneered computational frameworks to map and analyze dynamic three-dimensional electric fields in proteins.
- Revealed how local electric fields correlate with reaction outcomes, enabling *mechanistic insights* and guiding *rational enzyme design*.

Enzyme Design Through Electric Fields & Second Coordination Spheres

- Showcased the potential to rewire enzyme reactivity by tuning protein electric fields, paving the way for custom biocatalysts.
- Demonstrated how second coordination sphere and long-range interactions modulate hydrogen-atom transfer and reaction specificity.

Hybrid QM/MM Reaction Mechanisms

- Executed adiabatic mapping and free energy simulations (metadynamics, umbrella sampling) to dissect complex enzyme pathways.
- Employed spin natural orbital analysis to connect active-site electronic structure with *metalloenzyme catalytic efficiency*.

Classical Molecular Dynamics & Protein Motion Analysis

- Developed force field parameters for metalloenzymes and nucleic acids, capturing metal-ligand coordination accurately.
- Applied principal component analysis and dynamic cross-correlation to reveal long-range motions underlying enzyme function.

High-Performance Computing & Software Management

- Administered UNIX-based clusters, handling software installations and system optimization.
- Oversaw computational campaigns, scaling simulations to high-throughput levels and ensuring reproducibility.

PEER-REVIEWED PUBLICATIONS

(23) Eberhart, M. E., Alexandrova, A. N., Ajmera, P., Bim. D., Chaturvedi, S. S., Vargas, S., Wilson, T. R. "Methods for Theoretical Treatment of Local Fields in Proteins and Enzymes" **Chem. Rev.** 2025. https://doi.org/10.1021/acs.chemrev.4c00471.

(22) Ajmera, P., & Vargas, S., & **Chaturvedi, S. S.**, Hennefarth, M., Alexandrova, A. N. "PyCPET-Computing Heterogeneous 3D Protein Electric Fields and Their Dynamics" 2024, https://doi.org/10.26434/chemrxiv-2024-r3thp.

(21) Vargas, S., & **Chaturvedi, S. S.,** & Alexandrova, A. N. "Machine-Learning Prediction of Protein Function from the Portrait of Its Intramolecular Electric Field" ***J. Am. Chem. Soc.*** 2024, *146*(41), 28375–28383.

(20) Thomas, M. G., & Simahudeen, B. JSR., & **Chaturvedi, S. S.**, Gorantla, K. R., White, W., # Wildey, J., # Schofield, C. J., Christov, C. Z. "The Unique Role of the Second Coordination Sphere to Unlock and Control Catalysis in Nonheme Fe(II)/2-Oxoglutarate Histone Demethylase KDM2A" ***Inorg. Chem.*** 2024, *63*(23), 10737–10755.

(19) **Chaturvedi, S. S.**, & Vargas, S., & Ajmera, P., & Alexandrova, A. N. "Directed Evolution of Protoglobin Optimizes the Enzyme Electric Field" ***J. Am. Chem. Soc.*** 2024, *146*(24), 16670–16680.

(18) Varghese, A., & Waheed, S. O., & **Chaturvedi, S. S.**, DiCastri, I., # LaRouche, Ciara., # Kasi, B., # Lehnert, N., Li, D., Christov, C. Z., Karabencheva-Christova, T. G. "Revealing the Catalytic Strategy of FTO" ***Chem Catalysis*** 2023, *3*, 100732.

(17) **Chaturvedi, S. S.**, Bim, D., Christov, C. Z., Alexandrova, A. N. "From Random to Rational: Improving Enzyme Design Through Electric Fields, Second Coordination Sphere Interactions, and Conformational Dynamics" ***Chem. Sci.*** 2023, *14*, 10997–11011.

(16) Simahudeen, B. JSR., & **Chaturvedi, S. S.**, Warner, C., # Wildey, J., # White, W., # Thompson, M., Schofield, C. J., Christov, C. Z. "Catalysis by KDM6 Histone Demethylases – A Synergy Between the Non-Heme Iron(II) Center, Second Coordination Sphere, and Long-Range Interactions" ***Chem. Eur. J.*** 2023, *29*(51), e202301305. **(Selected for the front cover).**

(15) **Chaturvedi, S. S.**, Simahudeen, B. JSR., & Ramanan, R., Rankin, J. A., Hu, J., Hausinger, R. P., Christov, C. Z. "Can an External Electric Field Switch Between Ethylene Formation and L-Arginine Hydroxylation in the Ethylene Forming Enzyme?" ***Phys. Chem. Chem. Phys.*** 2023, *25*, 13772–13783.

(14) **Chaturvedi, S. S.**, Thomas, M. G., & Wildey, J., # Warner, C., # White, W., # Schofield, C. J., Hu, J., Hausinger, R. P., Karabencheva-Christova, T. G., Christov, C. Z. "Dioxygen Binding Is Controlled by the Protein Environment in Non Heme Fe(II) and 2-Oxoglutarate Oxygenases: A Study on Histone Demethylase PHF8 and an Ethylene-Forming Enzyme" ***Chem. Eur. J.*** 2023, *29*(24), e202300138.

(13) **Chaturvedi, S. S.**, Simahudeen, B. JSR., & Waheed, S. O., & Wildey, J., # Warner, C., # Schofield, C. J., Karabencheva-Christova, T. G., Christov, C. Z. "Can Second Coordination Sphere and Long-Range Interactions Modulate Hydrogen Atom Transfer in a Non-Heme Fe(II)-Dependent Histone Demethylase?" ***JACS Au*** 2022, *2*(9), 2169–2186.

(12) Waheed, S. O., & Varghese, A., & **Chaturvedi, S. S.**, Karabencheva-Christova, T. G., Christov, C. Z. "How Human TET2 Enzyme Catalyzes the Oxidation of Unnatural Cytosine Modifications in Double-Stranded DNA" ***ACS Catal.*** 2022, *12*(9), 5327–5344.

(11) Varghese, A., & **Chaturvedi, S. S.**, DiCastri, B., # Mehler, E., # Fields, G. B., Karabencheva-Christova, T. G. "Effects of the Nature of Metal Ion, Protein and Substrate on the Catalytic Center in Matrix Metalloproteinase-1: Insights from Multilevel MD, QM/MM and QM Studies" ***ChemPhysChem.*** 2021, *23*(4), e202100680. (Selected for the front cover and cover profile).

(10) Varghese, A., & **Chaturvedi, S. S.**, Fields, G. B., Karabencheva-Christova, T. G. "A Synergy Between the Catalytic and Structural Zn(II) Ions and the Enzyme and Substrate Dynamics Underlies the Structure–Function Relationships of Matrix Metalloproteinase Collagenolysis" ***J. Biol. Inorg. Chem.*** 2021, *26*(5), 583–597.

(9) Waheed, S. O., & **Chaturvedi, S. S.**, Karabencheva-Christova, T. G., Christov, C. Z. "Catalytic Mechanism of Human Ten-Eleven Translocation-2 (TET2) Enzyme: Effects of Conformational Changes, Electric Field, and Mutations" ***ACS Catal.*** 2021, *11*(7), 3877–3890.

(8) **Chaturvedi, S. S.**, Ramanan, R., Hu, J., Hausinger, R. P., Christov, C. Z. "Atomic and Electronic Structure Determinants Distinguish Between Ethylene Formation and l-Arginine Hydroxylation Reaction Mechanisms in the Ethylene-Forming Enzyme" ***ACS Catal.*** 2021, *11*(3), 1578–1592.

(7) Waheed, S. O., & Ramanan, R., **Chaturvedi, S. S.**, Lehnert, N., Schofield, C. J., Christov, C. Z., Karabencheva Christova, T. G. "Role of Structural Dynamics in Selectivity and Mechanism of Non-Heme Fe(II) and 2-Oxoglutarate Dependent Oxygenases Involved in DNA Repair" ***ACS Cent. Sci.*** 2020, *6*(5), 795–814. (Selected for the supplementary cover image).

(6) Ramanan, R., & **Chaturvedi, S. S.**, Lehnert, N., Schofield, C. J., Karabencheva-Christova, T. G., Christov, C. Z. "Catalysis by the JmjC Histone Demethylase KDM4A Integrates Substrate Dynamics, Correlated Motions, and Molecular Orbital Control" ***Chem. Sci.*** 2020, *11*(36), 9950–9961.

(5) **Chaturvedi, S. S.**, Ramanan, R., Lehnert, N., Schofield, C. J., Karabencheva-Christova, T. G., Christov, C. Z. "Catalysis by the Non-Heme Iron (II) Histone Demethylase PHF8 Involves Iron Center Rearrangement and Conformational Modulation of Substrate Orientation" ***ACS Catal.*** 2020, *10*(2), 1195–1209.

(4) **Chaturvedi, S. S.**, Ramanan, R., Waheed, S. O., & Karabencheva-Christova, T. G., Christov, C. Z. "Structure–Function Relationships in KDM7 Histone Demethylases" ***Adv. Protein Chem. Struct. Biol.*** 2019, *117*, 113–125.

(3) **Chaturvedi, S. S.**, Ramanan, R., Waheed, S. O., & Ainsley, J., & Evison, M., Ames, J. M., Schofield, C. J., Karabencheva-Christova, T. G., Christov, C. Z. "Conformational Dynamics Underlies Different Functions of Human KDM7 Histone Demethylases" ***Chem. Eur. J.*** 2019, *25*(21), 5422–5426.

(2) Waheed, S. O., & Ramanan, R., **Chaturvedi, S. S.**, Ainsley, J., & Evison, M., Ames, J. M., Schofield, C. J., Karabencheva-Christova, T. G., Christov, C. Z. "Conformational Flexibility Influences Structure–Function Relationships in Nucleic Acid N-Methyl Demethylases" ***Org. Biomol. Chem.*** 2019, *17*, 2223–2231.

(1) Ainsley, J., & **Chaturvedi, S. S.**, Karabencheva-Christova, T. G., Tanasova, M., Christov, C. Z. "Integrating Molecular Probes and Molecular Dynamics to Reveal Binding Modes of GLUT5 Activatory and Inhibitory Ligands" ***Chem. Commun.*** 2018, *54*, 9917–9920. (Equal contribution with the first author).

& Graduate student, # Undergraduate student

CONFERENCE PRESENTATIONS

Oral Presentations

Chaturvedi, S. S., Vargas, S., & Ajmera, P., & Alexandrova, A. N. "Directed Evolution Optimizes Electric Field in Protoglobin for Non-Native Catalysis" Gordon Research Seminar - Bioinorganic Chemistry, 2024, Ventura, California.

Chaturvedi, S. S., Vargas, S., & Alexandrova, A. N. "Directed Evolution Optimizes Electric Field in Protoglobin for Non-Native Catalysis" SoCal TheoChem 2023, 2023, University of California, Los Angeles, California.

Chaturvedi, S. S., Hu, J., Hausinger, R. P., Christov, C. Z. "Factors influencing ethylene production in the ethylene-forming enzyme" *ACS National Meeting*, Fall 2022, Bioinorganic Chemistry Session, Virtual.

Poster Presentations

Chaturvedi, S. S., Vargas, S., & Alexandrova, A. N. "Electric Fields as Key Determinants of Enzyme Catalysis" SoCal Theo Chem 2024, 2024, University of California, San Diego, California.

Chaturvedi, S. S., Vargas, S., & Alexandrova, A. N. "Directed Evolution Optimizes Electric Field in Protoglobin for Non-Native Catalysis" Gordon Research Conference - Metals in Biology, 2024, Ventura, California.

Chaturvedi, S. S., Wildey, J., # Warner, C., # Karabencheva-Christova, T. G., Christov, C. Z. "Second Coordination Sphere and Long-Range Interactions as Tools for Modulating HAT in a Histone Demethylase", *ACS National Meeting*, Fall 2022, Bioinorganic Chemistry Session, Virtual.

Chaturvedi, S. S., Wildey, J., # Warner, C., # Karabencheva-Christova, T. G., Christov, C. Z. "Iron Center Rearrangement in Catalytic Mechanism of Human Non-Heme Iron (II) Histone Demethylase PHF8", *ACS UPLS*, Fall 2021, Marquette, USA.

Chaturvedi, S. S., Ramanan, R., Hu, J., Hausinger, R. P., Christov, C. Z. "Structural Determinants for ethylene Formation and L-Arginine Hydroxylation Reaction Mechanism in the Ethylene-Forming Enzyme", *ACS National Meeting*, Fall 2021, Bioinorganic Chemistry Session, Virtual.

Chaturvedi, S. S., Ramanan, R., Karabencheva-Christova, T. G., Christov, C. Z. "Computation Insights into Catalytic Mechanism of Nε-Methyl Lysine Demethylation by PHF8", *Michigan Technological University*, Department of Chemistry Poster Session, 2019, Houghton, USA.

Chaturvedi, S. S., Ramanan, R., Karabencheva-Christova, T. G., Christov, C. Z. "Computation Insights into Catalytic Mechanism of Nε-Methyl Lysine Demethylation by PHF8", *ACS National Meeting*, Fall 2019, Sci-Mix Session, San Diego, USA.

Chaturvedi, S. S., Ramanan, R., Karabencheva-Christova, T. G., Christov, C. Z. "Computation Insights into Catalytic Mechanism of Nε-Methyl Lysine Demethylation by PHF8", *ACS National Meeting*, Fall 2019, Bioinorganic Chemistry Session, San Diego, USA.

& Graduate student, # Undergraduate student

TEACHING

CH3521 Physical Chemistry Lab II
Graduate Teaching Assistant, Michigan Technological University — Spring 2019

CH3511 Physical Chemistry Lab I
Graduate Teaching Assistant, Michigan Technological University — Fall 2018

CH1151 University Chemistry Lab I
Graduate Teaching Assistant, Michigan Technological University — Spring 2018

MENTORING & TRAINING

As a Postdoctoral Scholar (University of California, Los Angeles, 2023–Present)

- **Graduate Research Mentorship**: Provide hands-on training in electric-field analysis, QM/MM reaction mechanism calculations, molecular dynamics simulations, and machine learning approaches for new graduate students joining the Alexandrova Lab. Emphasize rigorous computational workflows and reproducible data practices.
- **Leadership in Collaborative Projects**: Coordinated and help supervise multiple projects within the lab's bio-subgroup, ensuring research aims are well-aligned and junior researchers receive clear guidance. Provide targeted feedback on data analysis and presentation skills, enhancing each project's overall impact at group meetings, seminars, and conferences.
- **Publication & Grant Support**: Assist graduate and undergraduate students in writing manuscripts, responding to peer reviews, and applying for fellowships or computational grants.
- **Ongoing Career Guidance**: Advise mentees on research ideas/proposals, postgraduate plans, and academic networking strategies, leveraging insights from my own transition from Ph.D. student to postdoc.

During Graduate Studies (Michigan Technological University, 2018–2022)

- **New Graduate Student Onboarding:** Served as a senior graduate student mentor, training three incoming Ph.D. students in force field parameter development, molecular dynamics, and QM/MM simulations. Guided them through setting up HPC jobs, analyzing mechanistic pathways, and interpreting data.
- **Undergraduate Research Engagement:** Mentored five undergraduates on computational modeling and molecular dynamics simulations, collaborating on research that yielded four co-authored publications and two poster presentations at ACS conferences.
- **Interdisciplinary Collaboration:** Helped junior lab members connect with experimental collaborators (Michigan State, Oxford) to validate computational findings, thus improving their communication skills and ability to integrate experimental data.
- **Professional Skill Development:** Shared experiences on effective manuscript writing, conference presentations, and best practices for time management and lab record-keeping, resulting in enhanced productivity and publication output for the group.

RESEARCH GRANTS

Maximize ACCESS CHE160054 Co-PI: Shobhit S. Chaturvedi
$Estimated Value of Awarded Resources: $121,706, University of California, Los Angeles — 2025

Maximize ACCESS CHE160054 Co-PI: Shobhit S. Chaturvedi
$Estimated Value of Awarded Resources: $36,522, University of California, Los Angeles — 2024

Discover ACCESS CHE240077 PI: Shobhit S. Chaturvedi
$Estimated Value of Awarded Resources: $14,838, University of California, Los Angeles — 2024

Discover ACCESS CHE240053 PI: Shobhit S. Chaturvedi

$Estimated Value of Awarded Resources: $14,838, University of California, Los Angeles 2024

Discover ACCESS BIO230147 PI: Shobhit S. Chaturvedi
$Estimated Value of Awarded Resources: $14,838, University of California, Los Angeles 2023

Discover ACCESS CHE230072 PI: Shobhit S. Chaturvedi
$Estimated Value of Awarded Resources: $14,838, University of California, Los Angeles 2023

Discover ACCESS CHE230069 PI: Shobhit S. Chaturvedi
$Estimated Value of Awarded Resources: $14,838, University of California, Los Angeles 2023

AWARDS & FELLOWSHIPS

Poster Presentation Award
SoCal Theo Chem 2024, University of California, San Diego 2024

Doctoral Finishing Fellowship
Michigan Technological University 2022

Health Research Institute Fellowship
Michigan Technological University 2022

Poster Presentation Award
1st Place, ACS Upper Peninsula Local Section 2021

Career Enrichment Grant
Graduate Student Government, Michigan Technological University 2020 & 2021

Outstanding Graduate Student Summer Research Award
Department of Chemistry, Michigan Technological University 2020

Travel Grant
Graduate Student Government, Michigan Technological University 2019

AAAS/Science Program for Excellence in Science Award
American Association for the Advancement of Science 2018

Rotary Youth Leadership Award
Rotary Club of Nagpur 2016

Project Exhibition Award
3rd Place, NCOAT-NIRMITI, Priyadarshini Institute of Engineering & Technology 2016

Oral Presentation Award
1st Place, Unnati 15, Priyadarshini College of Engineering 2015

INVOLVEMENT

Chemistry Department Representative
Graduate Student Government, Michigan Technological University September 2018 - May 2019

Member of Academic Committee
Graduate Student Government, Michigan Technological University — September 2018 - May 2019

Secretary
Rotaract Club, Priyadarshini Institute of Engineering and Technology — June 2015 - May 2016

Head of Abhyudaya Board
Department of Chemical Engineering, Priyadarshini Institute of Engineering and Technology — June 2015 - May 2016

PROFESSIONAL AFFILIATIONS

American Chemical Society — June 2019 - Present

American Association for Advancement of Science — August 2018 - Present